

7. Computation as a 3-way Relation

The structure

Computation is a relation between a **machine** M , a **system of representation** R , and a **function** f .

Given any two, you can solve for the third.

$$M + R = f?$$

$$M + f = R?$$

$$f + R = M?$$

Most of the problems I will assign are of the third kind.

The three elements are logically independent. (R)

The same M can compute different functions f and f^* relative to different systems of representation R and R^* .

The same function f can be computed by different machines M and M^* relative to different systems of representation R and R^* .

(Section on Representation: choice of representational system matters. Tie in to Newell)